Heralded Single photon sources through Spontaneous Four Wave Mixing

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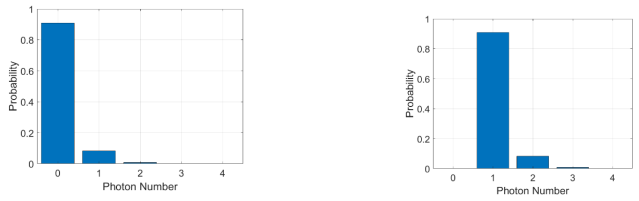
***Abstract*—In this project, we evaluate the design of high-purity single photon sources, specifically discussing a non-linear method called Spontaneous Four Wave Mixing (SFWM) in silicon photonics. In-order to achieve a purity of above 93%, we discuss a design which uses asymmetric ring resonator. 3D FDTD simulations were done for the asymmetric ring resonator and results are evaluated in $1.55\ \mu m$, validating the design for high-purity quantum applications.**

I. Methodology

We can generate single photon sources via two methods:deterministic photon gun and probabilistic heralded photon source. In photon gun, we make a quantum dot which has a discrete energy levels in a very small spatial configuration allowing only one electron to get excited. When the electron is excited, the photon gun emits a photon. This is made with III-V semiconductor as Si is an indirect band gap semiconductor. One another disadvantage of this method is that, it is difficult to integrate it into a PIC as there will be a lot of coupling losses.On the other hand, Heralding photon source is based on temporal correlation between a pair of photons generated by a non-linear process (SFWM in our case). As SFWM forces the electrons to be generated in pair, detection of one photon (we call it idler) will ensure the presence of another photon (we call this signal). This conditional generation of photon is termed as heralded. The probabilistic nature comes from here. It is given by the equation,

$$P(n) = \frac{\mu^n}{(1+\mu)^{n+1}}$$

To overcome this probabilistic nature, we can use multiplexing techniques like spatial,temporal and frequency multiplexing.



(a) Normal photon source (b) Heralded photon source

Fig. 1: Comparison of typical photon sources: (a) normal single-photon source and (b) heralded single-photon source

The above plot shows how photon distribution in normal and heralded sources differ. The heralded photon sources acts as a single photon source for the most of the time.

II. Spontaneous Four Wave Mixing

Due to the Kerr effect the refractive index will depend on $\|E\|^2$. This results in generation of other frequencies when a light of one frequency is passed through. For example, when a pair pump photons of frequency ω_p is sent through a waveguide, there will be generation of idler and signal photons of frequency ω_i and ω_s that will satisfy energy conservation via $\omega_p + \omega_p = \omega_i + \omega_s$. For a continuous pump source, ω_p is almost constant implying that we can track the generation of ω_i and ω_s photons. The above statement implies that the signal and idler photons are **entangled**!! This is very bad for our generated signal photon as we need 100% pure photon as the output of single photon source. This can be solved by sending a pulsed laser as pump where a photon of frequency ω_{p_1} and another photon of frequency ω_{p_2} can coexist simultaneously generating signal and idler photons by satisfying the condition $\omega_{p_1} + \omega_{p_2} = \omega_i + \omega_s$. As ω_{p_1} and ω_{p_2} can take a wide range of values, the signal and idler photons wont be 100% entangled now. We define a quantity called as Joint Spectral Amplitude (JSA) which quantifies the purity of the photons. It is defined by the following equation,

$$\begin{aligned} F(\omega_s, \omega_i) = \int d\omega_p\, \alpha(\omega_p)\alpha(\omega_s + \omega_i - \omega_p) \\ \times\, l_p(\omega_p) l_p(\omega_s + \omega_i - \omega_p) l_s(\omega_s) l_i(\omega_i) \end{aligned} \tag{1}$$

where $\alpha(\omega_p)$ and $\alpha(\omega_s + \omega_i - \omega_p)$ are the spectral amplitude of the generated pump photons which is desired to be wide as discussed above and $l_p(\omega_p)$, $l_p(\omega_s + \omega_i - \omega_p)$, $l_s(\omega_s)$ and $l_i(\omega_i)$ are the frequency characteristics of the device that we are considering to generate SFWM. The purity can be calculated form JSA by taking Schmidt decomposition along the orthonormal basis of signal and idler basis states as shown below.

$$F(\omega_s, \omega_i) = \sum_n \sqrt{\lambda_n} u_n(\omega_s) v_n(\omega_i) \tag{2}$$

The purity P is given by $P = \sum_n \lambda_n^2$. Therefore for 100% purity($P = 1$), we need $\lambda_1 = 1$ with JSA being,

$$F(\omega_s, \omega_i) = u(\omega_s) v(\omega_i) \tag{3}$$

Now let us use Micro-ring resonator to generate SFWM and estimate the purity.

III. Micro-ring Resonators

We know from coupled mode theory that, when two waveguides are close to each other, the evanescent tail of one waveguide interacts with other resulting in coupling between the two waveguides. We can replace the top waveguide with a ring to get resonance for specific wavelengths. The following picture shows the schematic of an ideal micro-ring symmetric where both bus and ring have same width.

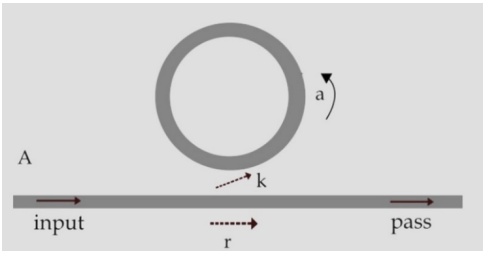


Fig. 2: Micro-ring resonator

The following plot shows the frequency response of the micro-ring resonator of $a = 0.98$ and $r = 0.79$.

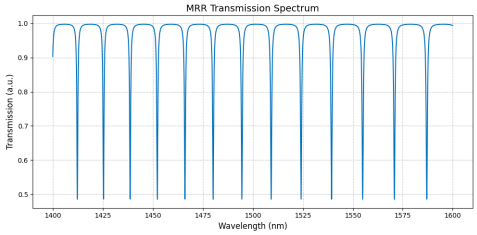


Fig. 3: MRR frequency response

From the equation 1 and equation 3, we infer that to get high purity ($\lambda = 1$), we need $\alpha(\omega_p)$ and $\alpha(\omega_s + \omega_i - \omega_p)$ to be almost constant over the ω_p. Therefore the Q-factor which measures the width of the resonance frequency must be lower for pump compared to that of signal and idler for us to get high purity of signal. As we can see from the frequency response of symmetric MRR, it is impossible to find three resonant frequencies where $Q_p \ll Q_s\ ,\ Q_i$. The following plot shows the JSA plot and purity measured from the (1) and (3) for two extreme cases.

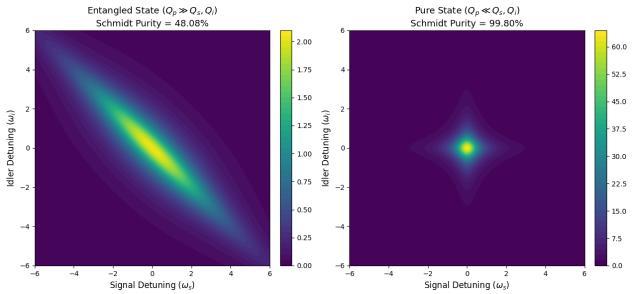


Fig. 4: JSA plot

In order to achieve this feat, we can use different widths for bus and ring. This helps us vary the transmission percentage due to the phase mismatch as well. We will now discuss the 3D-FDTD simulation for symmetric and asymmetric waveguides.

IV. FDTD Simulations

In-order to avoid high computational cost, instead of using an actual ring we use a rectangular waveguide of length L_{dc} and width. We put a source inside the bus waveguide and a monitor inside the ring and find the cross-coupling coefficient. This will help us decide the pump, signal and idler frequencies with the condition $Q_p \ll Q_s\ ,\ Q_i$. We know that the Q factor and cross coupling coefficient (T) are inversely related. From the T vs wavelength plot, we need to choose pump wavelength as the one with the maxima and signal and idler wavelength as minima.

A. Symmetric waveguide

The following table shows the parameters of the symmetric waveguide simulated in the 3-FDTD.

TABLE I: Symmetric Micro-Ring Resonator Parameters

Parameter	**Value**
Length of bus	70 μm
Width of bus	500 nm
Length of ring	60 μm
Width of ring	500 nm
Seperation distance	50 nm
Source position from center	32 μm
Monitor position from center	28 μm

B. Symmetric waveguide

The following table shows the parameters of the symmetric waveguide simulated in the 3-FDTD.

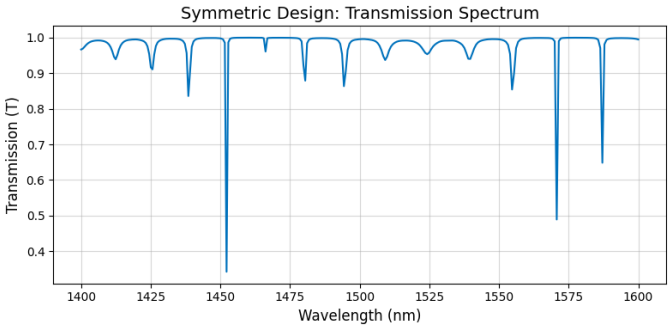
TABLE II: Aymmetric Micro-Ring Resonator Parameters

Parameter	**Value**
Length of bus	70 μm
Width of bus	500 nm
Length of ring	60 μm
Width of ring	480 nm
Seperation distance	50 nm
Source position from center	32 μm
Monitor position from center	28 μm

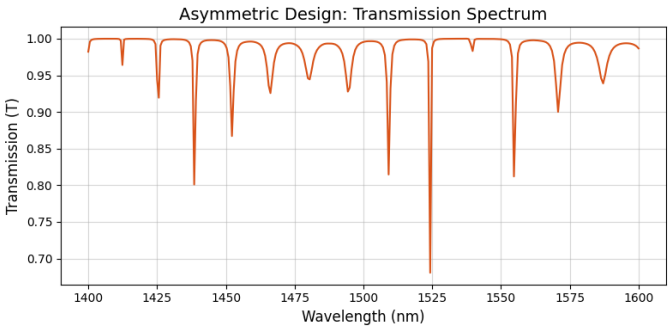
V. Results

The results of the fdtd are generated in a .dat file which can be found here. We got a purity of 91.2 % for symmetrical waveguide and a purity of 95.5% for asymmetrical waveguide.

The following picture shows the transmission spectrum of symmetrical and asymmetrical waveguides.



(a) Symmetrical waveguide



(b) Asymmetrical waveguide

From the transmission spectrum one can see that if we choose 1525 nm as the pump, the spectrum is wider than the adjacent spectrums which can be considered for signal and idler. Due to the phase mismatch, the primary maxima is shifted from around 1450 nm to 1525 nm.